

U.S. Department of Health and Human Services
Assistant Secretary for Technology Policy and the Office of the National Coordinator for Health
Information Technology
330 C Street SE
Washington, D.C. 20201

**Subject: HHS Health Sector AI RFI - Request for Information: Accelerating the Adoption
and Use of Artificial Intelligence as Part of Clinical Care**

To Whom it May Concern:

We appreciate the opportunity to provide, on behalf of Penn Medicine, a response to the RFI on Accelerating the Adoption and Use of Artificial Intelligence (AI) as Part of Clinical Care. Penn Medicine is building a future-ready health system where AI augments human expertise, improves patient outcomes, accelerates research, and reduces administrative burden, and we appreciate HHS' attention to the importance of this work.

Penn Medicine is one of the world's leading academic medical centers, dedicated to the related missions of medical education, biomedical research, and excellence in patient care. Penn Medicine consists of the Raymond and Ruth Perelman School of Medicine at the University of Pennsylvania (founded in 1765 as the nation's first medical school) and the University of Pennsylvania Health System.

Penn Medicine cares for patients in facilities and their homes, stretching from the Susquehanna River in Pennsylvania to the New Jersey shore. Our facilities include the Hospital of the University of Pennsylvania, Penn Presbyterian Medical Center, Chester County Hospital, Lancaster General Health, Princeton Health, and Pennsylvania Hospital—the nation's first hospital, chartered in 1751. Additional facilities and enterprises include Penn Medicine at Home, Good Shepard Penn Partners Rehabilitation, Lancaster Behavioral Health Hospital, and Princeton House Behavioral Health, among others.

1. What are the biggest barriers to private sector innovation in AI for health care and its adoption and use in clinical care?

Trust – Healthcare delivery organizations, clinicians, and patients/consumers need to develop trust in health AI tools and their developers, particularly when lives are on the line. This means that we need to be able to develop evidence of safety and effectiveness of tools, particularly in high-risk circumstances closer to the patient/during direct patient care. Then, we need to be able to monitor the performance of tools once implemented to ensure continued safety and effectiveness, an effort that becomes more complex as multiple AI tools operate together. This will take deep partnership between health AI developers and implementers throughout the life cycle of a tool. Healthcare moves at the speed of trust, and tools will not be adopted/used without that trust. Of course, need to keep in mind what the evidentiary standard should be – we are aiming for health AI tools not to be perfect but as good as or better than our current practice.

Evidence – The vast majority of published scientific manuscripts about clinical AI tools, and the vast majority of FDA-authorized AI-enabled devices, have no evidence to suggest that they improve clinical decisions, care processes, or patient outcomes. A key element of trust (as described above) is developing this evidence so that clinicians, patients, and health system leaders are well informed when making decisions about purchasing and deployment. Developing this evidence also creates an opportunity for private sector innovators to distinguish their products from those that have less (or no) evidence of clinical effectiveness. Furthermore, such evidence would alleviate critical information asymmetries in investment decisions for clinical AI systems that have so far led to significant market inefficiencies. Evidence development can take at least two forms including (1) experimental frameworks for generating the evidence with lower risk and cost; (2) standardized and routine monitoring, as distinct from up front, one-time evaluation which is what is prevalent. Of note, even when evidence is present, currently much of it relates to foundation model performance in isolation of real workflows. This is not equivalent to performance or attributed outcomes in real-world clinical environments.

Uncertainty around legal, regulatory, and medical malpractice issues – There is significant uncertainty around the way health AI will be legislated and regulated which can inhibit innovation as well as adoption of the technology. For example, it is unclear the direction in which federal and state policy will go in relation to health AI, which one will have a more dominant effect, and which of these stakeholders vendor-partners and healthcare organizations should engage with in discussions relevant to the future of health AI. Companies may hesitate to enter a market in which they must navigate 50 potentially divergent state regulatory regimes rather than a single coherent framework. Additionally, ongoing uncertainty around which CDS tools meet the definition of a device (despite recent attempts at clarification), combined with recent instances of the FDA exhibiting “enforcement discretion” without transparent rationale or defined timelines, further deters companies from entering the market. Finally, concerns of clinicians and vendors related to malpractice and liability resulting from AI-generated information or actions are currently unresolved.

Interoperability and data standardization - Data are the lifeblood of health AI, and tools need to both be able to access data from systems across the healthcare ecosystem (patients/consumers, clinicians, payers, pharmacies, etc.) but then also be able to take actions flexibly within environments like the electronic health record and within clinician workflows (functional in addition to technical interoperability). Importantly, data standardization and adoption of standards such as FHIR, along with expanded data element availability, across broad ranges of use cases is critical to truly enabling transformative health AI use cases.

FHIR and other APIs should also be enabled with appropriate capabilities to write back into the EHR and integrate deeply within clinician and patient workflows to make it easier for clinicians and patients to use health AI tools.

Finally, there should be cost controls on integrations and API access.

All of this, of course, needs to be balanced by adequate privacy and information security safeguards.

Competency/workforce development and process reimagination – In order to derive the most benefit from health AI tools, individuals within organizations must learn how to utilize tools effectively and process must be modified or reimagined entirely to make use of the health AI tool

Business model/reimbursement – Aside from a few specific technologies (ie AI guided coronary plaque analyses), AI-based tools are not (and may not be) reimbursed for directly, making it even more important for vendor-partners to clearly articulate and measure outcomes and return on investment from health AI tools. CMS’ and the FDA’s paired ACCESS/TEMPO models offer a limited initial framework, but without resolving fundamental questions about how the FDA will regulate many AI tools beyond temporary enforcement discretion, it remains unclear how these efforts will scale beyond pilot programs.

An effective business model is also important as AI implementers shoulder impactful AI-related costs, such as those arising from (1) liability risks (which bear unknown costs at the moment) and (2) operating costs from both model costs and data integration costs. Many implementers can't realistically experiment, let alone adapt and reimagine people and process along with technology, without financial relief on these fronts.

Of note, much of the innovation in the health AI space is focused on the use of large language models, and that focus could become a barrier in the future to newer innovations. HHS should consider encouraging divergent thinking in health AI and continuing to explore other forms of language, vision-language, world models, among other AI tools and methods.

2. What regulatory, payment policy, or programmatic design changes should HHS prioritize to incentivize the effective use of AI in clinical care and why? What HHS regulations, policies, or programs could be revisited to augment your ability to develop or use AI in clinical care? Please provide specific changes and applicable Code of Federal Regulations citations.

Information blocking– Enforcement of 21st Century Cures Act information blocking provisions would help to create data ecosystem needed for effective health AI development and implementation and the environment needed to implement health AI tools flexibly deeply within workflows.

Interoperability – Similarly, continued development of trusted interoperability frameworks such as TEFCA and CMS-Aligned Networks will allow for health AI tools to be more effectively implemented in workflow .

Virtual care – In addition to information/insights generated from health AI tools, patients/consumers also need access to care delivery in order to improve health outcomes, and that virtual care is an excellent pairing with health AI tools such as chatbots. However, virtual care, in the form of synchronous (ambulatory visits, Hospital at Home), asynchronous care (eConsults, eVisits), and remote monitoring needs to have permanent reimbursement to ensure stability needed to develop and grow programs. Relatedly, variation in state licensure and telehealth regulations governing interstate delivery of care creates additional barriers to scaling AI-enabled care models.

Clarification of intention for oversight – Informal discussions with FDA officials, in addition to the precedent set by the recent *ACLA vs FDA* Supreme Court case, all suggest that “homegrown” AI models – or those that are developed and deployed locally by hospitals but not marketed or sold – are not likely to be considered part of the FDA’s oversight purview. Clarifying this point officially would help to alleviate uncertainty in regulatory and business decisions and would help foster local AI innovation.

Incentives – Similar to other eras of health IT implementation (for example post-HITECH Act), tying incentives, particularly financial incentives to effective adoption and utilization of

health AI to boost patient outcomes will drive adoption. Efforts such as CMMI's ACCESS model and solicitations such as ARPA-H's ADVOCATE are a step in the right direction.

Liability – Concerns around liability limit more enthusiastic use of health AI in clinical care delivery. Now, most if not all liability is placed on the shoulders of implementers of health AI rather than developers. To encourage implementation, HHS/CMS should consider how this risk can be shared more equally between implementers and developers.

3. For non-medical devices, we understand that use of AI in clinical care may raise novel legal and implementation issues that challenge existing governance and accountability structures (e.g., relating to liability, indemnification, privacy, and security). What novel legal and implementation issues exist and what role, if any, should HHS play to help address them?

There are numerous legal and implementation issues raised in the context of health AI implementation that challenge existing structures. A key example of this is the case of AI agents. AI agents are already beginning to interact with each other to impact care delivery, including teams of agents working within a healthcare delivery system as well as agents interacting between actors such as payers and clinicians.

Especially when these interactions extend beyond one entity (such as payer and clinician), HHS should consider setting standards for how these interactions should occur. This has already started to happen via, for example, the CMS Interoperability and Prior Authorization Final Rule (encouraging adoption of FHIR standards for data exchange) as well as recent work on real-time prescription benefits – but this approach likely needs to continue to evolve in an era of agentic interactions.

In the clinician space, systems are under development where agents review the medical literature, or large bodies of medical records, and surface insights and recommendations which can serve as the basis for patient facing actions (i.e. orders, communication) by other agents. As we mention above, this will require new frameworks for health AI evaluation as well as liability shared between health AI developers and implementers.

From an HHS perspective, it will be important to balance creating conditions for safe and effective implementation of health AI spanning patients, clinicians, payers, developers and others while leaving room for experimentation.

4. For non-medical devices, what are the most promising AI evaluation methods (pre- and post-deployment), metrics, robustness testing, and other workflow and human-centered evaluation methods for clinical care? Should HHS further support these processes? If so, which mechanisms would be most impactful (e.g., contracts, grants, cooperative agreements, and/or prize competitions)?

There are a number of AI evaluation methods emerging that fall into several categories. These include (but are not limited to):

Benchmarks to evaluate health AI tools applied in specific task

areas (ie summarization): There are emerging constructs such as MedHELM and HealthBench which show promise in the evaluation of health AI particularly pre-deployment.

Bedi S, Cui H, Fuentes M, et al. Holistic evaluation of large language models for medical tasks with MedHELM. *Nat Med*. Published online January 20, 2026. doi:[10.1038/s41591-025-04151-2](https://doi.org/10.1038/s41591-025-04151-2)

Automated evaluation of AI output: There are also promising models of LLM-as-judge to evaluate artifacts such as AI-generated summaries at scale, including post-deployment.

Croxford E, Gao Y, First E, et al. Evaluating clinical AI summaries with large language models as judges. *npj Digit Med*. 2025;8(1):640. doi:[10.1038/s41746-025-02005-2](https://doi.org/10.1038/s41746-025-02005-2)

Chung P, Swaminathan A, Goodell AJ, et al. Verifying Facts in Patient Care Documents Generated by Large Language Models Using Electronic Health Records. *NEJM AI*. 2025;3(1). doi:[10.1056/AIdbp2500418](https://doi.org/10.1056/AIdbp2500418)

Evaluation of the performance of compound AI systems (ie multiple models working together): Much more work needs to be done in this space, especially as there is very little known about evaluation of single and multi-agent systems pre- or post-deployment, including how agents may be able to supervise each other, and how humans might maintain appropriate oversight in a human on the loop model.

HHS should continue to encourage pragmatic and applied research activity in this space. Programs like ARPA-H ADVOCATE are also encouraged as a way to spur government, academic, and industry partnership.

Phases of research – While phases of research are well established for the evaluation of pharmaceutical and traditional medical devices, such a framework has yet to be widely applied to AI-enabled medical devices. Thus, established frameworks offer an opportunity to foster innovation while guiding the appropriate interpretation of evidence for AI systems. HHS should support enhanced evaluation alongside the development of AI-specific evaluation methods and frameworks across all phases of research. For example, the vast majority of evidence for AI-enabled clinical systems is best characterized as Phase 0, yet the results of such studies are often over-interpreted and misunderstood. More Phase 1 and 2 studies are needed to bridge knowledge gaps that would support more high quality, Phase 3 randomized evaluations of AI systems that are deployed in real clinical workflows.

Park Y, Jackson GP, Foreman MA, Gruen D, Hu J, Das AK. Evaluating artificial intelligence in medicine: phases of clinical research. *JAMIA Open* 2020;3(3):326–31.

Support from HHS is critical here in all of the forms proposed including grants and contracts.

5. How can HHS best support private sector activities (e.g., accreditation, certification, industry-driven testing, and credentialing) to promote innovative and effective AI use in clinical care?

We believe new models need to emerge for credentialing health AI tools, potentially based on existing structures such as CLIA and current forms of medical licensing. This would help to build trust in health AI tools, and would enable adoption of health AI tools not requiring humans in every loop, while still maintaining human accountability. Two papers below from Penn faculty describe these models:

Herman DS, Reece JT, Weissman GE. Lessons for local oversight of AI in medicine from the regulation of clinical laboratory testing. *npj Digit Med*. 2024;7(1):359. doi:[10.1038/s41746-024-01369-1](https://doi.org/10.1038/s41746-024-01369-1)

Bressman E, Shachar C, Stern AD, Mehrotra A. Software as a Medical Practitioner— Is It Time to License Artificial Intelligence? Published online 2025.

On the other hand, allowing states to advance autonomous AI use cases without clear federal credentialing or oversight frameworks risks creating regulatory fragmentation and may undermine trust, accountability, and consistency in how these tools are deployed.

As noted above, HHS should fund cooperative research and development efforts in this space and use its authority to create innovation payment models (such as ACCESS) and regulatory sandboxes for safe and effective experimentation.

6. Where have AI tools deployed in clinical care met or exceeded performance and cost expectations and where have they fallen short? What kinds of novel AI tools would have the greatest potential to improve health care outcomes, give new insights on quality, and help reduce costs?

In care delivery, health AI has already had positive, measurable impact, particularly in reducing friction in healthcare for patients, clinicians, and administrators.

Clinical documentation – Ambient AI scribes used to augment clinician documentation have been shown to reduce time in chart as well as to reduce clinician burnout and increase professional satisfaction. See other parts of our response for supporting evidence.

Closing care gaps – Generative AI based tools are able to review the fragmented medical record and identify incidentally diagnosed or missed findings (such as pulmonary nodules which can become cancer) and connect patients to care at earlier, more treatable stages of disease (or prevent disease from developing entirely).

Upskilling the healthcare workforce – AI powered tools are able to bring what was previously specialized knowledge to the point of care of all clinicians, allowing more clinicians to participate in the delivery of that care, and to augment patient access to care.

Korom R, Kiptinness S, Adan N, et al. Ai-based clinical decision support for primary care: a real-world study. *arXiv*. Preprint posted online 2025.
doi:10.48550/ARXIV.2507.16947

Predicting risk of disease – Predictive AI based tools are able to use electronic health record data to predict clinical events such as inpatient decompensation, or risk of mortality, and connect clinicians and patients to appropriate downstream actions.

Rossetti SC, Dykes PC, Knaplund C, et al. Real-time surveillance system for patient deterioration: a pragmatic cluster-randomized controlled trial. *Nat Med*. 2025;31(6):1895-1902. doi:[10.1038/s41591-025-03609-7](https://doi.org/10.1038/s41591-025-03609-7)

Courtright KR, Madden V, Bayes B, et al. Default Palliative Care Consultation for Seriously Ill Hospitalized Patients: A Pragmatic Cluster Randomized Trial. *JAMA*. 2024;331(3):224. doi:[10.1001/jama.2023.25092](https://doi.org/10.1001/jama.2023.25092)

Clinical imaging – Clinical imaging use cases such as AI-assisted mammography have led to improved patient outcomes

Lång K, Josefsson V, Larsson AM, et al. Artificial intelligence-supported screen reading versus standard double reading in the Mammography Screening with Artificial Intelligence trial (MASAI): a clinical safety analysis of a randomised, controlled, non-inferiority, single-blinded, screening accuracy study. *The Lancet Oncology*. 2023;24(8):936-944. doi:[10.1016/S1470-2045\(23\)00298-X](https://doi.org/10.1016/S1470-2045(23)00298-X)

Thus far, health AI tools have mainly provided information, but novel tools will also be able to reason, use tools, plan, have memory and take action to augment humans in healthcare delivery (agents).

There is also early empirical evidence that implementation of AI tools may degrade clinicians' independent diagnostic and decision-making skills ("deskilling"). Greater attention and research investment are needed to better understand, monitor, and mitigate these risks.

Budzyń K, Romańczyk M, Kitala D, Kołodziej P, Bugajski M, Adami HO, Blom J, Buszkiewicz M, Halvorsen N, Hassan C, Romańczyk T. Endoscopist deskilling risk after exposure to artificial intelligence in colonoscopy: a multicentre, observational study. *The Lancet Gastroenterology & Hepatology*. 2025 Oct 1;10(10):896-903.

7. Which role(s), decision maker(s), or governing bodies within health care organizations have the most influence on the adoption of AI for clinical care? What are the primary administrative hurdles to the adoption of AI in clinical care?

Within healthcare organizations, CEOs and other executive leaders help to create the vision for use of health AI within an organization. Technology executives such as Chief Digital and Information Officers, Chief Data and Analytics Officers, Chief Health AI Officers, and Chief Nursing Information Officers, and Chief Health Information Officers are charged with strategy and tactics behind implementing health AI in care delivery. This is often done in collaboration with medical and nursing leaders such as Chief Physician Executives, Chief Medical Officers, Chief Clinical Officers, and Chief Nursing Officers.

Most healthcare delivery organizations also now have health AI governance committees with multidisciplinary stakeholders across teams in the organization (medical, legal, IT, ethics, etc.) which also help to guide decision-making around health AI. Even with these committees, a significant hurdle is risk assessment, and without unifying standards or certifications, each assessment of each health AI tools is ad hoc. Simplifying the risk assessment process would speed adoption of health AI in clinical care.

Outside of healthcare organizations – medical specialty societies have tremendous influence on the behavior of medical practitioners and should be engaged in developing, implementing, and monitoring health AI.

See above on commentary regarding administrative hurdles to adoption.

8. Where would enhanced interoperability widen market opportunities, fuel research, and accelerate the development of AI for clinical care? Please consider specific data types, data standards, and benchmarking tools.

Enhanced interoperability, through growth of trusted interoperability frameworks such as TEFCA, adoption and expansion of FHIR-based standards, promotion of activities such as real-

time prescription benefit sharing, and prevention of information blocking, among other actions, is critical for nurturing innovation in health technology. Importantly, interoperability should include interoperability frameworks, data exchange standards, content standards, and workflow standards to be truly effective.

These actions and others promote healthy competition amongst vendor-partners, ultimately bringing the best deeply workflow integrated applications to market, and allowing systems to craft technology environments that more rapidly bring the best applications to benefit patients and consumers.

9. What challenges within health care do patients and caregivers wish to see addressed by the adoption and use of AI in clinical care? Equally, what concerns do patients and caregivers have related to the adoption and use of AI in clinical care?

Patients and caregivers are seeking assistance across their healthcare journeys, starting from accessing information on known and potential health conditions, contextualized for them, to being connected to care at the right time in the right place, appropriately preparing for encounters, having a productive encounter with a clinician, navigating the care delivery system to take action from encounters, obtaining and understanding testing, and understanding and navigating health benefits.

Both patients/caregivers and clinicians share concerns around implementation of health AI, starting with being able to trust outputs of health AI tools. Healthcare delivery is intrinsically a human endeavor, and patients, caregivers, and clinicians alike are concerned AI may negatively impact the human relationships, judgement, reasoning, and empathy that are at the core of healthcare delivery. There are also concerns related to the environmental impact of AI (water and energy use), the impact of data centers, the privacy and security of protected health information, the impact of AI on jobs in the healthcare and other industries, and consent and notification of the use of AI in care delivery environments.

10. Are there specific areas of AI research that HHS should prioritize to accelerate the adoption of AI as part of clinical care?

There is much to be learned about the evaluation and implementation of health AI which would result in accelerated adoption including

Evaluation and ongoing monitoring of health AI: This is a critical need to ensure safety and effectiveness of tools. This needs to apply to predictive AI, generative AI, single agent AI, and multi-agent AI systems.

Development of safety and control systems for health AI: Whether agentic or otherwise, health AI systems, especially when semi-autonomous or autonomous, will need mechanisms to ensure behaviors remain aligned with human values.

Human skill development: As mentioned in this response, the use of health AI is starting to be associated with erosion in human skills. Research is needed on which human skills in healthcare delivery truly continue to be needed, and how to maintain and enhance those skills while collaborating with AI.

Use of health AI across tasks: In order for health AI to be most effective, it must also be able to handle multiple related tasks together that would fall under a job. It is unknown across a range of healthcare jobs, clinical and nonclinical, how best to extend beyond using AI for a specific

task (ie documenting a single note) to a job (i.e. completing set of tasks for an ambulatory visit, from preparation through followup).

Health AI implementation and behavioral science: In order for health AI to translate to improved patient outcomes, tools must be effectively implemented and used. There is research needed in how to ensure our healthcare delivery workforce develops competency in using health AI. Strategies to nudge use of health AI may come from adaptations of learnings from the field of behavioral science, both for clinicians and patients/consumers.

Human-computer interaction: We need to better understand how to shape interactions between humans and AI (human computer collaboration) to generate positive outcomes, including how to architect interactions where humans are in the loop (preventing automation bias and neglect) and on the loop. This applies to patient and clinicians but particularly to patients, as illustrated in a recent RCT.

Bean AM, Payne RE, Parsons G, et al. Reliability of LLMs as medical assistants for the general public: a randomized preregistered study. *Nat Med*. Published online February 9, 2026. doi:[10.1038/s41591-025-04074-y](https://doi.org/10.1038/s41591-025-04074-y)

Part of this work needs to be on how humans and computers share work, which models work and adapt best for certain jobs/tasks (i.e. centaur vs. cyborg).

Abdulnour REE, Gin B, Boscardin CK. Educational Strategies for Clinical Supervision of Artificial Intelligence Use. *N Engl J Med*. 2025;393(8):786-797. doi:[10.1056/NEJMr2503232](https://doi.org/10.1056/NEJMr2503232)

a. *Are there published findings about the impact of adopted AI tools and their use clinical care?*

There is now a large body of literature around the impact of ambient generative AI powered documentation tools, including literature from Penn. Namely, the tools have been found to save clinicians time, save cognitive load, save time in chart, and significantly improve clinician experience. This has been seen in observational studies as well as randomized controlled trials.

Duggan MJ, Gervase J, Schoenbaum A, et al. Clinician Experiences With Ambient Scribe Technology to Assist With Documentation Burden and Efficiency. *JAMA Netw Open*. 2025;8(2):e2460637. doi:[10.1001/jamanetworkopen.2024.60637](https://doi.org/10.1001/jamanetworkopen.2024.60637)

Afshar M, Ryan Baumann M, Resnik F, et al. A Pragmatic Randomized Controlled Trial of Ambient Artificial Intelligence to Improve Health Practitioner Well-Being. *NEJM AI*. 2025;2(12). doi:[10.1056/AIoa2500945](https://doi.org/10.1056/AIoa2500945)

Lukac PJ, Turner W, Vangala S, et al. Ambient AI Scribes in Clinical Practice: A Randomized Trial. *NEJM AI*. 2025;2(12). doi:[10.1056/AIoa2501000](https://doi.org/10.1056/AIoa2501000)

Additionally, there is evidence that summarization of data in the EHR can be generated effectively and safely by large language models.

Small WR, Austrian J, O'Donnell L, et al. Evaluating Hospital Course Summarization by an Electronic Health Record–Based Large Language Model. *JAMA Netw Open*. 2025;8(8):e2526339. doi:[10.1001/jamanetworkopen.2025.26339](https://doi.org/10.1001/jamanetworkopen.2025.26339)

There is also now a body of literature on the impact of health AI tools on augmenting patient experience and clinician decision making.

Zeltzer D, Kugler Z, Hayat L, et al. Comparison of initial artificial intelligence (Ai) and final physician recommendations in ai-assisted virtual urgent care visits. *Ann Intern Med*. 2025;178(4):498-506. doi:10.7326/ANNALS-24-03283

O'Sullivan JW, Palepu A, Saab K, et al. A large language model for complex cardiology care. *Nat Med*. Published online February 6, 2026. doi:[10.1038/s41591-025-04190-9](https://doi.org/10.1038/s41591-025-04190-9)

Tu T, Schaekermann M, Palepu A, et al. Towards conversational diagnostic artificial intelligence. *Nature*. 2025;642(8067):442-450. doi:[10.1038/s41586-025-08866-7](https://doi.org/10.1038/s41586-025-08866-7)

See above for more research findings.

Penn Medicine appreciates the opportunity to comment on RFI and welcomes any questions you may have.

Sincerely,

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